

A project report on

Stress Analysis of a 3-Dimension Block

For the course of Stress analysis in Mechanical Design MECH-6441 of Master of Engineering in
Mechanical Engineering

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Project scope

The scope of this project is to find out the displacement distribution using the Pascal pyramid from sets of the polynomial equations at the given coordinates of the block in terms of the X, Y and Z using Numpy.

The stress distributions and strain distributions also determined from these equations.

To find stress distribution as per given condition, two different elasticity matrix are used.

From stress and strain tensor, principal stresses and strains and their directions, octahedral stresses and maximum shear stresses at the corner points are determined using Numpy function.

To check the failure criterion, Von Mises failure criterion approach was used to find which corner on the top face of the block has failed by material yielding.

Chapter 1: Problem Statement

Consider a block of size 30 mm x 20 mm x 10 mm made of two metallic materials. The bottom half of the block is made of Aluminum and the top half is made of Steel. The displacements of 10 material points of the block have been measured after certain loading is applied, and they are given in the Table below.

- Determine the distributions (i.e. as functions of coordinates of material points) of displacements, strains and stresses in the block, in the xyz co-ordinate system. Use suitable material properties for steel and aluminum.
- Determine the principal stresses, their directions, maximum shear stress, and the octahedral stresses at each mid-face points of the block.
- For the chosen materials, using Von Mises failure criterion and approximation, determine which corner on the top face of the block has failed by material yielding.
- Calculate the % change in the angle at the top-right corner on the front face of the block.

Table 1.1 Co-ordinates of the block before and after loading

Points	Co-ordinates before loading			Co-ordinates after loading		
	X	Y	Z	X _l	Y _l	Z _l
A	5	0	20	5.003	0.002	19.004
B	30	0	20	30.003	0.0	19.995
C	30	10	20	29.995	10.003	19.996
D	0	10	20	0.0	10.009	19.995
E	0	0	0	0.0	0.2	0.0
F	30	0	0	30.004	0.002	0.004
G	30	10	0	29.996	9.995	0.3
H	0	10	0	0.0011	9.995	0.0
I	30	5	10	29.987	5.005	9.986
J	11	9	8	11.009	9.009	8.001

Chapter 2: Introduction

In this part, a brief introduction of the geometry is discussed along with the properties of the materials used.

The geometry was designed in Solid works with the top half to be steel and the bottom half is aluminium to understand the problem statement better.

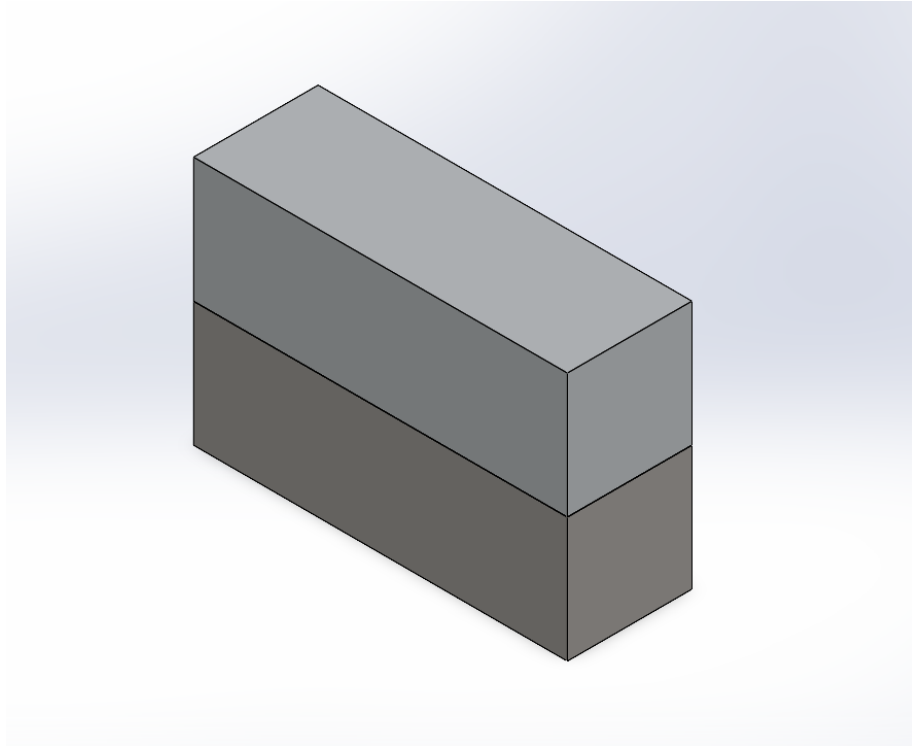


Figure 2.1 Geometry

Table 2.1 Material Property of Steel and Aluminum

Material	Steel	Al 6061-T6
Modulus of Elasticity, E	210×10^3 MPa	69×10^3 MPa
Poisson's Ratio, ν	0.30	0.33
Shear Modulus, G	79.3×10^3 MPa	26×10^3 MPa
Yield Strength, σ_{sy}	215 MPa	110 MPa

Chapter 3: Solution I

We will discuss about the Pascal Pyramid, equation generation using Pascal pyramid and distribution of the displacement, strain and stress of the block.

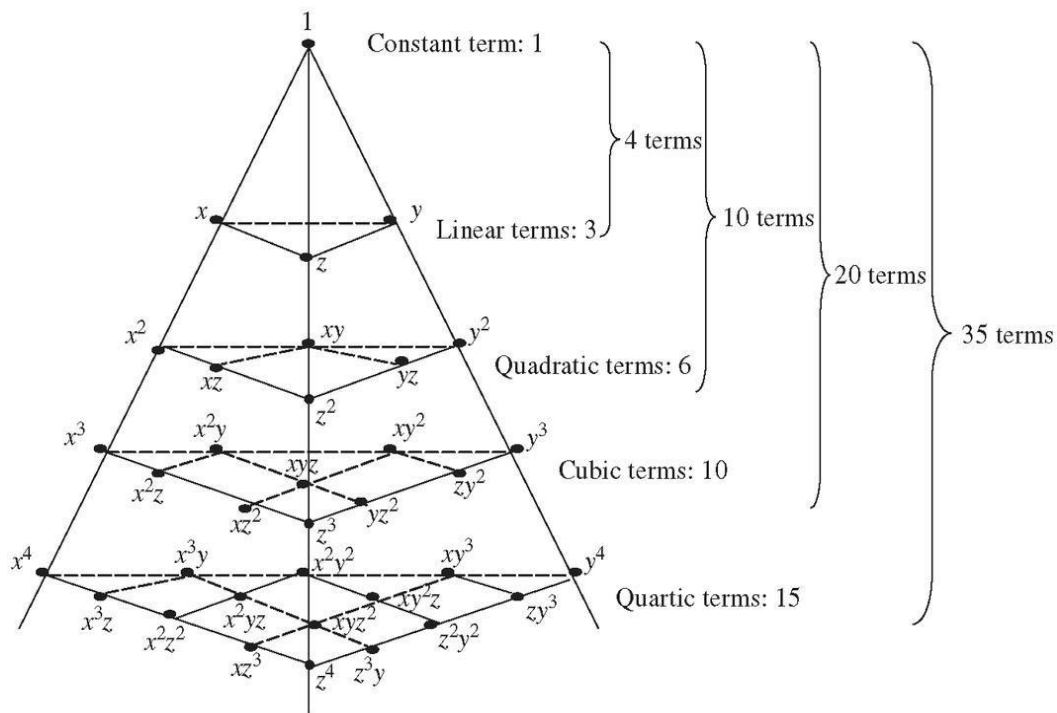
3.1 Pascal Pyramid

From the problem statement we can see that we have a total of 10 coordinates. In order to formulate the mathematical equation using the given 10 points which satisfies our requirements, we use the 3D Pascal Pyramid to formulate the equation.

However, there are a few procedures to follow while using the Pascal pyramid to formulate the displacement in the X, Y, Z direction.

- I. It should start from the top layer and come down in sequence.
- II. Terms can be skipped, given the reason is valid. Skipping terms will not affect the outcome.
- III. Axis of symmetry should be followed.
- IV. We will eliminate other polynomial possibility to choose the accurate polynomial formulation with reference to the maximum shear stress.

Figure 3.1 Pascal Pyramid [4]



We can depict the pyramid as a table which can be used to formulate the equation for maximum shear. We will eliminate the other polynomial equations after we get that equation.

Sr No.	Constant	Linear			Quadratic						Cubic			Quartic		
1	1	x	y	z	xy	yz	zx	x ²	y ²	z ²						
2	1	x	y	z	xy	yz	zx				xyz	x ² y	xy ²			
3	1	x	y	z	xy	yz	zx				xyz	x ² z	zy ²			
4	1	x	y	z	xy	yz	zx				xyz	xz ²	yz ²			
5	1	x	y	z	xy	yz	zx				x ³	y ³	z ³			
6	1	x	y	z	xy	yz	zx				xyz			xyz ²	x ² y ²	
7	1	x	y	z	xy	yz	zx				xyz			x ² yz	xy ² z	
8	1	x	y	z	xy	yz	zx							xyz ²	x ² yz	xy ² z
9	1	x	y	z	xy	yz	zx							xyz ²	xz ³	z ³ y
10	1	x	y	z	xy	yz	zx							xyz ²	x ² z ²	z ² y ²

Table 3.1 Possibility of Polynomial Set from Pascal Pyramid

Using the above table, we found the displacement equations in the X, Y, Z directions to be,

$$U = a_1 + a_2x + a_3y + a_4z + a_5x^2 + a_6y^2 + a_7z^2 + a_8xy + a_9yz + a_{10}zx$$

$$V = b_1 + b_2x + b_3y + b_4z + b_5x^2 + b_6y^2 + b_7z^2 + b_8xy + b_9yz + b_{10}zx$$

$$W = c_1 + c_2x + c_3y + c_4z + c_5x^2 + c_6y^2 + c_7z^2 + c_8xy + c_9yz + c_{10}zx$$

After getting the values from all the polynomials, we get the more realistic values at the points given in the problem statement for U, V and W.

3.2 Displacement Distribution

By using that polynomial equation set we get the displacement distributions in terms of x, y and z. The coefficients of the above equations are found using the Numpy program and substituted.

$$\text{Coefficient} = [\text{eqn_mat}]^{-1} * [\text{Displacement}]$$

Displacement Component along the X-Direction,

$$U = 0 + 0.000953333x - 0.010729333y + 0.0028646667z - 2.73333 \times 10^{-5}x^2 + 1.0839333 \times 10^{-3}y^2$$

$$-1.459833 \times 10^{-4} z^2 - 3.03333 \times 10^{-5} xy + 1.6667 \times 10^{-7} xz.$$

Displacement Component along the Y-Direction,

$$V = 0.200 - 0.04692x - 0.2021906667y + 0.0920453333z + 0.001344x^2 + 0.0181690667y^2 - 0.0045922667z^2 + 0.00066xy - 0.00001xz + 0.00005yz.$$

Displacement Component along the Z-Direction,

$$Z = -0 + (-0.2967066667)x + (-1.392088)y + 0.692994z + 0.0098946667x^2 + 0.1392088y^2 - 0.0339247z^2 + 0.0009866667xy - 0.0004983333xz - 0.001475yz.$$

3.3 Strain Distribution

Partial distribution of the partial derivative of U,V and W will give the strain distribution in terms of X,Y and Z

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x} ; \quad \varepsilon_{yy} = \frac{\partial u_y}{\partial y} ; \quad \varepsilon_{zz} = \frac{\partial u_z}{\partial z}$$

$$\gamma_{xy} = \alpha + \beta = \partial u_x / \partial y + \partial u_y / \partial x$$

$$\gamma_{yz} = \partial u_y / \partial z + \partial u_z / \partial y$$

$$\gamma_{zx} = \partial u_x / \partial z + \partial u_z / \partial x$$

Figure 3.2 Strain Equation

Using the above set of equations, the values are found to be,

$$\mathbf{E}_{xx} = 0.0009533333 - 5.46666 \times 10^{-5} x - 3.03333 \times 10^{-5} y + 1.667 \times 10^{-7} z.$$

$$\mathbf{E}_{yy} = -0.2021906667 + 0.00066x + 0.0363381334y + 0.00005z.$$

$$\mathbf{E}_{zz} = 0.692994 - 0.0004983333x - 0.001475y - 0.0678494z.$$

$$\mathbf{G}_{xy} = -0.0576493333 + 0.0026576667x + 0.0028278666y - 0.00001z.$$

$$\mathbf{G}_{yz} = -1.3000426667 + 0.0009766667x + 0.2784676y - 0.0106595334z.$$

$$\mathbf{G}_{zx} = -0.293842 + 0.0197895001x + 0.0009866667y - 0.0007902999z.$$

3.4 Stress Distribution

We can relate the strain with the stress with the help of Hooke's law. The stress strain relation is as follows,

Here D is the elasticity matrix and it is calculated using the material properties.

$$\{\sigma\} = [D]\{\varepsilon\}$$

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{bmatrix} = \begin{bmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{bmatrix}$$

$$\mu = G = E/[2(1 + \nu)]$$

$$\lambda = \frac{\nu E}{(1 + \nu)(1 - 2\nu)}$$

Figure 3.3 Stress-Strain Relationship

Stress Distribution function for Aluminium

$$\sigma_{xx} = 24815.6 + 2.551X + 1752.3Y - 3413.59Z$$

$$\sigma_{yy} = 14279.7 + 39.646X + 3640.43Y - 3411.01Z$$

$$\sigma_{zz} = 60716.2 - 20.465X + 1678.43Y - 6934.5Z$$

$$\tau_{xy} = -1495.2 + 68.915X + 73.285Y - 0.25940Z$$

$$\tau_{yz} = -33722.8 + 25.344X + 7220.90Y - 276.46Z$$

$$\tau_{xz} = -7618.1 + 513.2X + 25.583Y - 20.504Z$$

Stress Distribution function for Steel

$$\sigma_{xx} = 59747.6 + 4.133X + 4215Y - 8219.57Z$$

$$\sigma_{yy} = 26897.4 + 119.58X + 10093.5Y - 8211.52Z$$

$$\sigma_{zz} = 171595.6 - 67.53X + 3981.7Y + 19199Z$$

$$\tau_{xy} = -4654.2 + 214.6X + 228.6Y - 0.8077Z$$

$$\tau_{yz} = -105000 + 78.88X + 22500Y - 861.2Z$$

$$\tau_{xz} = -23743 + 1598.5X + 79.7Y - 63.8Z$$

Chapter 4: Solution II

Here we will discuss about the principal stresses and strains, their directions. We will also discuss about the maximum shear stress and octahedral shear stress at each corner point of the block made of aluminium and steel.

Since the block is made up of half Aluminium and half steel, the material of the corner points will be of 2 different types, They are listed below in the table.

Point	Material	Coordinate Before Loading		
E	Aluminium	0	0	0
F	Aluminium	30	0	0
G	Aluminium	30	10	0
H	Aluminium	0	10	0
T1	Steel	0	0	20
B	Steel	30	0	20
C	Steel	30	10	20
D	Steel	0	10	20

Table 4.1 Coordinates of Corner Points of the Block

4.1 Principal Stresses, Their Directions and Maximum Shear Stresses

The maximum and minimum normal stresses at a point within a material, occurring on planes where there is no shear stress is called as principal stresses, and these planes are called principal planes. In this project we attain that from Numpy program. The eigen value of each stress tensor is demonstrated as below.

```
eigvals, eigvecs = np.linalg.eigh(sig)
```

$$\begin{aligned}
(\sigma_{xx} - \sigma)l_1 + \tau_{xy}m_1 + \tau_{xz}n_1 &= 0 \\
\tau_{yx}l_1 + (\sigma_{yy} - \sigma)m_1 + \tau_{yz}n_1 &= 0 \\
\tau_{zx}l_1 + \tau_{zy}m_1 + (\sigma_{zz} - \sigma)n_1 &= 0
\end{aligned}$$

$$\begin{vmatrix}
\sigma_{xx} - \sigma & \tau_{xy} & \tau_{xz} \\
\tau_{yx} & \sigma_{yy} - \sigma & \tau_{yz} \\
\tau_{zx} & \tau_{zy} & \sigma_{zz} - \sigma
\end{vmatrix} = 0$$

Which is re-written as

$$\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0$$

where

$$\begin{aligned}
I_1 &= \sigma_{xx} + \sigma_{yy} + \sigma_{zz} \\
I_2 &= \sigma_{xx}\sigma_{yy} + \sigma_{yy}\sigma_{zz} + \sigma_{zz}\sigma_{xx} - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{zx}^2 \\
I_3 &= \sigma_{xx}\sigma_{yy}\sigma_{zz} - \sigma_{xx}\tau_{yz}^2 - \sigma_{yy}\tau_{zx}^2 - \sigma_{zz}\tau_{xy}^2 + 2\tau_{xy}\tau_{yz}\tau_{zx}
\end{aligned}$$

Figure 4.1 Equation to calculate Principal Stress and its Direction

At any point subtraction of maximum and minimum value of principal stress with division by two will give the maximum shear stress.

$$[\sigma] = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

Where, $\sigma_1 > \sigma_2 > \sigma_3$

$$(\tau_{max})_{31} = \pm \frac{\sigma_3 - \sigma_1}{2}$$

Using all the above equations, the principal stresses and maximum shear stresses and their directions are formulated and tabulated below.

Points	E	F	G	H	T1	B	C	D
σ_1 (MPa)	79,115.569	78,406.846	106,404.636	105,515.487	-44,637.617	-44,922.327	25,711.726	21,145.843
σ_2 (MPa)	24,961.163	24,708.118	42,289.568	42,367.352	-104,067.517	-104,934.817	-62,634.048	-62,197.885
σ_3 (MPa)	-4,279.055	-2,665.845	22,445.008	22,604.932	-305,014.565	-302,177.304	-232,239.926	-229,795.457
$\tau_{\max}$ (MPa)	41697.31	40536.35	41979.81	41455.28	130188.47	128627.49	128975.83	125470.65

Table 4.2 Principal Stresses and Maximum Shear Stresses

	L1	M1	N1	L2	M2	N2	L3	M3	N3
E	0.99993	0.006667	0.007109	0.006669	0.99438	0.10488	0.007113	0.10488	-0.99449
F	0.99995	0.003733	0.007129	0.003734	0.99899	0.04446	0.007131	0.04446	-0.99901
G	0.99745	0.069746	0.008193	0.069778	0.97746	0.19699	0.008199	0.19699	-0.98042
H	0.99673	0.080083	0.006915	0.080124	0.97461	0.20905	0.006932	0.20905	-0.97786
T1	0.99558	0.089676	0.025106	0.089682	0.98173	0.16494	0.025122	0.16494	-0.98621
B	0.99593	0.068882	0.051855	0.068881	0.99509	0.08138	0.051896	0.08138	-0.99637
C	0.96876	0.24749	0.00377	0.24773	0.96881	0	0.004061	0	-0.99999
D	0.97529	0.22091	0.006406	0.22094	0.9753	0	0.006995	0	-0.99998

Table 4.3 Direction of Principal Stresses

4.2 Octahedral Stresses

A plane that makes the same angle with the three principal directions. Such a plane is called an "octahedral plane". The resultant normal stress and shear stresses acting on this plane are called as octahedral normal stress and octahedral shear stresses.

$$\tau_{oct} = \frac{[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]^{(1/2)}}{3}$$

$$\sigma_{oct} = (\sigma_1 + \sigma_2 + \sigma_3)/3$$

Figure 4.2 Equation for Octahedral Stresses

	E	F	G	H	T1	B	C	D
σ_{oct} (MPa)	33,265.892	33,483.040	57,046.404	56,829.257	-151,239.900	-150,678.149	-89,720.74	-90,282.50
τ_{oct} (MPa)	34,543.22	33,669.28	35,824.07	35,353.77	111,392.223	109,875.383	107,019.75	104,337.62

Table 4.4 Shear Stresses and Octahedral Stresses

4.3 Principal Strains and Their Directions

Same as the principal stress there exist three mutually orthogonal planes on which has no shear strains and only normal strains are present. These planes are called as principal planes with respect to strains and the corresponding normal strains are known as principal strains. Equations to calculate principal strains and their directions are given below. For this project the Numpy function "eigvals" is used.

$$\begin{bmatrix} 2(\varepsilon_{xx} - \varepsilon) & \gamma_{xy} & \gamma_{zx} \\ \gamma_{xy} & 2(\varepsilon_{yy} - \varepsilon) & \gamma_{yz} \\ \gamma_{xz} & \gamma_{yz} & 2(\varepsilon_{zz} - \varepsilon) \end{bmatrix} = 0$$

$$\varepsilon^3 - I_1\varepsilon^2 + I_2\varepsilon - I_3 = 0$$

where

$$I_1 = \varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}$$

$$I_2 = \varepsilon_{xx}\varepsilon_{yy} + \varepsilon_{yy}\varepsilon_{zz} + \varepsilon_{zz}\varepsilon_{xx} - (0.25)\gamma_{xy}^2 - (0.25)\gamma_{yz}^2 - (0.25)\gamma_{zx}^2$$

$$I_3 = \varepsilon_{xx}\varepsilon_{yy}\varepsilon_{zz} - 0.25[\varepsilon_{xx}\gamma_{yz}^2 + \varepsilon_{yy}\gamma_{zx}^2 + \varepsilon_{zz}\gamma_{xy}^2] + 0.25[\gamma_{xy}\gamma_{yz}\gamma_{zx}]$$

Figure 4.3 Set of Equations for Principal Strains

Using the above set of equations, we find the principal strains and their directions and they are formulated in tables below.

Corner	E1	E2	E3
E	1.047688	0.003842	-0.55977
F	1.030912	-0.00415	-0.53179
G	1.232496	-0.00335	-0.38586
H	1.218473	0.00127	-0.37966
T1	0.371843	0.003944	-1.24001
B	0.367673	-0.00383	-1.22486
C	0.543685	-0.00322	-1.05316

Table 4.5 Principal Strains

	L1	M1	N1	L2	M2	N2	L3	M3	N3
E	0.111329	0.45651	-0.88273	0.979554	-0.20019	0.020014	0.167572	0.866905	0.469462
F	0.122917	-0.45952	0.879623	0.980724	0.191905	-0.03679	0.151897	-0.86719	-0.47425
G	0.11296	0.582286	0.805098	0.975792	-0.21773	0.020563	0.187267	0.783286	-0.59279
H	0.101753	-0.57258	-0.81351	0.977208	0.210679	-0.02606	0.186308	-0.79232	0.580965
T1	0.188633	0.779378	-0.59748	0.975445	-0.21913	0.02212	0.113686	0.586985	0.801576
B	0.202015	-0.78584	0.584507	0.974347	0.221699	-0.03869	0.099183	-0.57733	-0.81047
C	0.167734	0.864293	0.474196	0.979456	-0.20072	0.01939	0.111941	0.461202	-0.88021

Table 4.6 Principal Strain's Direction

Chapter 5: Solution III

Now we will look at the Von Mises Failure criterion and approximation to determine if mid point of each edge on the rear face will fail by material yielding.

Von Mises criterion is a yield criterion in solid mechanics that predicts when a ductile material, such as a metal, will begin to deform plastically under a complex stress state. So, as given in the problem statement, we will determine which corner on the top face of the block has failed by material yielding.

$$\sigma_v = \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]}, \quad \tau_{\max} = \frac{\sigma_1 - \sigma_3}{2}.$$

Figure 5.1 Equation for Von Mises stress calculation

According to our material, all four top corners are steel, and the results are tabulated below,

Point	Coordinates (mm)	$\tau_{\max}$ (MPa)	Von-Mises σ_v (MPa)	Steel yield σ_y (MPa)	Status
T1	(0, 0, 20)	130,188.474	236,334.280	370	FAILED
B	(30, 0, 20)	128,627.489	233,116.088	370	FAILED
C	(30,10, 20)	128,975.826	227,057.465	370	FAILED
D	(0, 10, 20)	125,470.650	221,366.957	370	FAILED

Table 5.1 Maximum Shear Stress Failure Criteria

Chapter 6: Solution IV

Now we will discuss about the percentage change in the top right corner on the front face block.

We calculated the angle change in degree and attained percentage change using the the Numpy code. The line is given below.

```
angle_change_deg = np.degree(gamma_xz)
percent_change = 100 * angle_change_deg / 90.0
```

And the result is,

$$\Delta\theta \approx 0.5418^\circ$$
$$\approx 0.60209\%$$

Angle change = 0.5418°

And

The percentage change = 0.60209%

Chapter 7: Result and Discussion

This chapter we will compare the maximum shear stress and octahedral stress for our block from the results we got. They are depicted as bar graph below.

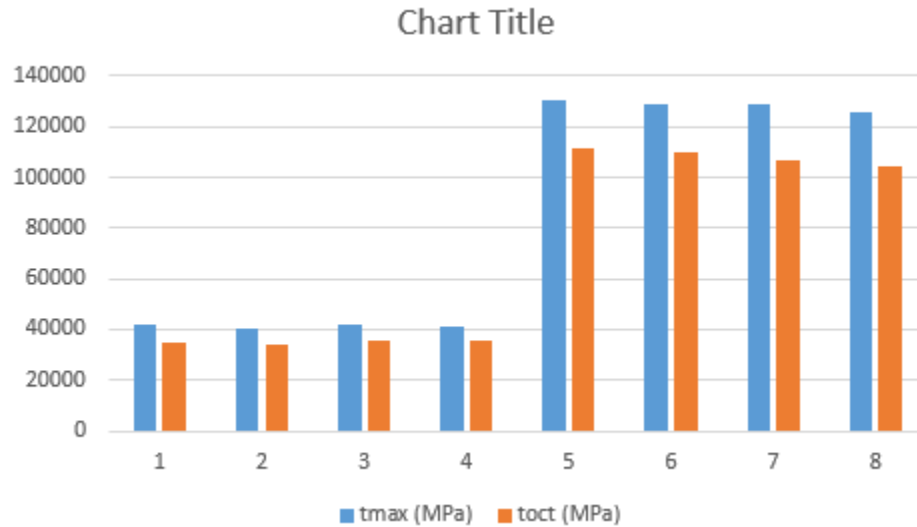


Figure 7.1 Comparison of Stresses

After looking at the comparison of the stresses, we can say stress on steel is much more than that of aluminium. It is almost three times as that of aluminium. It indicates that these corners of the block experiences lighter loading over all. The corners of steel experience higher loading in its corners. Point 5 has the highest value, closely followed by points 6, 7 and 8. These points are located on the top half of the block made by steel, it matches the expectation that steel carries most of the load.

Chapter 8: Conclusion

From the comparison done in the previous section signifies that aluminium in the bottom half having lower stiffness undergoes larger deformation, while the top half (steel) carries majority of the load. The Von-Mises criterion approach is more representative for ductile materials like the ones we have, indicating that steel begins to distort when the octahedral stress reaches almost one third of the materials yield strength.

Overall, the projects demonstrates that the top face with 4 points with steel only exceeds the Von-Mises threshold, thereby yielding locally and not affecting the other points of the block, especially aluminium not yielding and staying safe. This makes us realise the importance of failure conditions, Von-Mises in our case and this result is attained with the help of different polynomial equations and cleaver Numpy code. Thus proving its reliability and accuracy in helping us to our result and conclusion.

References

- [1] Dr. R. Ganesan Chapter 1 “STRESS” in Stress Analysis in Mechanical Design.
- [2] Dr. R. Ganesan Chapter 2 “STRAIN” in Stress Analysis in Mechanical Design.
- [3] Dr. R. Ganesan Chapter 3 “STRESS-STRAIN RELATION” in Stress Analysis in Mechanical Design.
- [4] <https://dalyfpascalstriangle.weebly.com/pascals-pyramid.html>
- [5] <https://www.simscale.com/docs/simwiki/fea-finite-element-analysis/what-is-von-mises-stress/>

Appendix

```
import numpy as np

points_before = np.array([
    [5, 0, 20],
    [30, 0, 20],
    [30, 10, 20],
    [0, 10, 20],
    [0, 0, 0],
    [30, 0, 0],
    [30, 10, 0],
    [0, 10, 0],
    [30, 5, 10]
], dtype=float)

points_after = np.array([
    [5.003, 0.002, 19.004],
    [30.003, 0.000, 19.995],
    [29.995, 10.003, 19.996],
    [0.000, 10.009, 19.995],
    [0.000, 0.200, 0.000],
    [30.004, 0.002, 0.004],
    [29.996, 9.995, 0.300],
    [0.0011, 9.995, 0.000],
    [29.987, 5.005, 9.986]
], dtype=float)

displacements = points_after - points_before

n = points_before.shape[0]
X = np.hstack([np.ones((n, 1)), points_before])
A_fit = np.linalg.lstsq(X, displacements, rcond=None)[0]

grad_u = A_fit[1:, :].T

strain = 0.5 * (grad_u + grad_u.T)

def hooke_stress(E, nu, eps):
    """Return stress tensor from strain tensor for isotropic
    material."""
    lam = (E * nu) / ((1 + nu) * (1 - 2 * nu))
    mu = E / (2 * (1 + nu))
    eps_vol = np.trace(eps)
```

```

        return lam * eps_vol * np.eye(3) + 2 * mu * eps

# Aluminum
E_Al, nu_Al, sigY_Al = 69000.0, 0.33, 276.0
sigma_Al = hooke_stress(E_Al, nu_Al, strain)

# Steel
E_St, nu_St, sigY_St = 210000.0, 0.30, 250.0
sigma_St = hooke_stress(E_St, nu_St, strain)

def stress_measures(sig):
    eigvals, eigvecs = np.linalg.eigh(sig)

    s_dev = sig - np.trace(sig)/3*np.eye(3)
    von_mises = np.sqrt(1.5 * np.sum(s_dev**2))

    max_shear = 0.5 * (np.max(eigvals) - np.min(eigvals))

    sigma_oct = np.trace(sig) / 3
    tau_oct = np.sqrt(2/3 * np.sum(s_dev**2))
    return eigvals, eigvecs, von_mises, max_shear, sigma_oct,
    tau_oct

vals_Al = stress_measures(sigma_Al)
vals_St = stress_measures(sigma_St)

gamma_xz = 2 * strain[0, 2]
angle_change_deg = np.degrees(gamma_xz)
percent_change = 100 * angle_change_deg / 90.0

np.set_printoptions(precision=6, suppress=True)

print("\nDisplacement gradient (du_i/dx_j):\n", grad_u)
print("\nSmall strain tensor:\n", strain)

print("\n--- ALUMINUM ---")
print("Stress tensor (MPa):\n", sigma_Al)
print("Principal stresses (MPa):", vals_Al[0])
print("Von Mises =", vals_Al[2], "MPa")
print("Max shear =", vals_Al[3], "MPa")
print("Octahedral normal =", vals_Al[4], "MPa")
print("Octahedral shear =", vals_Al[5], "MPa")

```

```
print("\n--- STEEL ---")
print("Stress tensor (MPa):\n", sigma_St)
print("Principal stresses (MPa):", vals_St[0])
print("Von Mises =", vals_St[2], "MPa")
print("Max shear =", vals_St[3], "MPa")
print("Octahedral normal =", vals_St[4], "MPa")
print("Octahedral shear =", vals_St[5], "MPa")

print("\nAngle change at top-right front corner (deg):",
angle_change_deg)
print("Percent change relative to 90°:", percent_change, "%")

print("\nYield check (Steel): Von Mises / Yield =", vals_St[2] /
sigY_St)
if vals_St[2] > sigY_St:
    print("=> Steel has yielded at top corners.")
else:
    print("=> Steel has NOT yielded at top corners.")
```